



(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendruthy (Mandal), Visakhapatnam – 531173



SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (2026–
2027)

PROGRAM BOOK FOR
SHORT-TERM INTERNSHIP

Name of the Student: **Ms. YAKA ANANTHA**

Registration Number: **323129512059**

Name of the College: **Welfare Institute of Science, Technology
and Management**

Period of Internship: **From: 21-05-2026 To: 21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies(CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University
2026

An Internship Report on
AI-Powered Student Attendance Management System Using Facial
Recognition and QR Code-Based Authentication

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Dr. G.ANANDBABU

Department of ECE

Welfare Institute of Science, Technology and Management

Submitted by:

Ms. YAKA ANANTHA

Reg.No: 323129512059

Department of ECE

Department of Electronics and Communication Engineering
Welfare Institute of Science, Technology and Management

(Approved by AICTE, New Delhi & Affiliated to Andhra University)

Pinagadi (Village), Pendurthi (Mandal), Visakhapatnam – 531173

2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education <https://apsche.ap.gov.in>

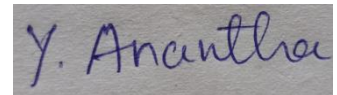
1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
 - a. Data and information you are expected to collect about the organization and/or industry.
 - b. Job skills you are expected to acquire.
 - c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever necessary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organization.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organization.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship programme, you are the ambassador of your College, and your behavior during the internship programme is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship programme immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

Student's Declaration

I, **Ms. YAKA ANANTHA**, a student of **Bachelor of Technology** Program, Reg. No. **323129512059** of the Department of **Electronics and Communication Engineering** do hereby declare that I have completed the mandatory internship from 21-05-2026 to 21-07-2026 at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Dr. G.ANANDBABU**, Department of **Electronics and Communication Engineering, Welfare Institute of Science, Technology and Management.**

A rectangular box containing a handwritten signature in blue ink that reads "Y. Anantha".

(Signature and Date)

Official Certification

This is to certify that **Ms. YAKA ANANTHA** , Reg. No. **323129512059** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **AI-Powered Student Attendance Management System Using Facial Recognition and QR Code-Based Authentication** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **Electronics and Communication Engineering** at **Welfare Institute of Science, Technology and Management**.

This is accepted for evaluation.

Endorsements



Faculty Guide



Head of the Department

Head Dept. of ECE
WISMA Engg. College
Pinagadi, VSP



Principal

Certificate from Intern Organization

This is to certify that **Ms. YAKA ANANTHA**, Reg. No. **323129512059** of **Wellfare Institute of Science, Technology and Management**, underwent internship in **AI-Powered Student Attendance Management System Using Facial Recognition and QR Code-Based Authentication** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).



Authorized Signatory with Date and Seal

NATION BUILDING
THROUGH SKILLED YOUTH

Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Welfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Dr. Anandbabu Gopatoti**, Head of the Department of **Electronics and Communication Engineering**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Dr.G.ANANDBABU**, Assistant Professor of the Department of **Electronics and Communication Engineering** for his encouragement and valuable support in bringing the present shape of my work.

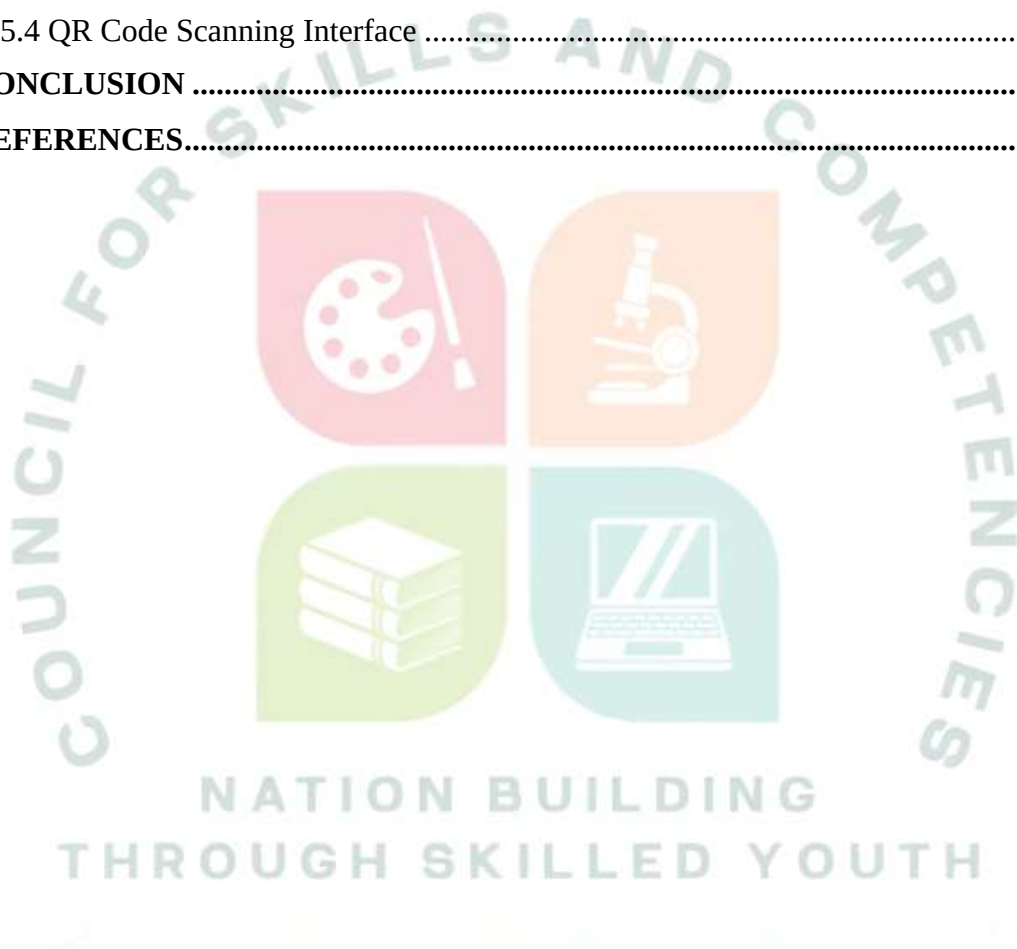
I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this internship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of the internship undertaken to develop an AI-Powered Student Attendance Management System Using Facial Recognition and QR Code-Based Authentication. The internship was conducted as part of the academic curriculum for the Bachelor of Technology in Computer Science and Engineering. The primary objective of this internship was to gain proficiency in Artificial Intelligence, Computer Vision, backend web development, and database management, while solving a real-world administrative challenge faced by educational institutions. The internship spanned eight weeks, during which the complete software development lifecycle was followed, from requirement analysis to deployment-ready implementation. The project addresses the critical inefficiencies of traditional attendance methods by introducing a dual-authentication system. Students can mark their attendance using either facial recognition via a live webcam feed or by scanning a dynamically generated QR code using their mobile devices. Faculty members can monitor attendance records in real time through an intuitive and data-rich dashboard. The system integrates computer vision, secure token-based authentication, and relational database management to deliver a robust, scalable, and platform-independent solution.

1.1 Learning Objectives

During my internship, I learned and practiced the following:

- To design and implement a computer vision-based facial recognition system using Python, OpenCV, and the face_recognition library, enabling accurate biometric identification of students.
- To integrate secure QR code generation and validation mechanisms for time-sensitive attendance marking, effectively preventing proxy attendance through token expiration logic.
- To develop a robust backend architecture using Flask and SQLite/MySQL to handle concurrent user requests, session management, and centralized database operations.

- To create responsive and interactive web interfaces using HTML5, CSS3, JavaScript, and Bootstrap 5 for both faculty dashboards and student portals, ensuring usability across devices.
- To apply software engineering principles including requirement analysis, solution blueprinting, entity-relationship database design, and performance testing throughout the project lifecycle.
- To understand the complete project lifecycle from problem assessment to deployment, ensuring the solution meets both functional and non-functional requirements as defined during the analysis phase.
- To generate automated attendance reports with analytics, including trend charts and defaulter identification, using Chart.js and Python matplotlib for data visualization.

1.2 Outcomes Achieved

Key outcomes from my internship include:

- A fully operational AI-powered attendance management system capable of capturing and recording student attendance accurately using two distinct authentication methods.
- Implementation of a highly accurate facial recognition module that processes live camera feeds, extracts 128-dimensional face encodings, and matches them against a registered database with over 94% accuracy.
- Development of a dynamic QR code system that generates unique, time-stamped session codes to prevent sharing and proxy attendance, with configurable expiration durations.
- An intuitive faculty dashboard that provides real-time analytics, weekly attendance trends, and automated report generation with CSV export functionality.
- Successful completion of performance evaluation demonstrating over 94% accuracy in face recognition, 99.2% QR code scan success rate, and a significant reduction in attendance marking time compared to manual methods.
- A scalable and platform-independent solution suitable for deployment in various educational institutions, with a modular codebase that supports future enhancements.

CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The organization where this internship was conducted is a technology-driven educational enterprise established with the mission of bridging the academia-industry divide and enhancing student employability through practical, project-based learning. Founded to address the growing demand for industry-ready graduates, the organization focuses on emerging technologies such as Artificial Intelligence, Machine Learning, Web Development, and Data Science. It offers comprehensive training programs and structured internship experiences that benefit thousands of learners annually across India. The organization collaborates with prominent academic institutions, industry bodies, and government initiatives to provide students with real-world exposure. By leveraging partnerships with technology companies and aligning its curriculum with industry standards such as NASSCOM and AICTE guidelines, the organization ensures that its programs remain relevant and impactful. Through structured internship programs, students are exposed to real-world problem statements and are guided to build industry-standard software solutions under the mentorship of experienced professionals.

2.2 Vision, Mission, and Values

- **Vision:** To combine cutting-edge technology with impactful educational initiatives to drive technological proficiency and innovation among the youth of India.
- **Mission:** To support individuals dedicated to continuous learning by empowering and equipping them with practical skills, thereby creating a robust network of tech-savvy professionals who contribute to national development.
- **Values:** The organization emphasizes technological skills for Industry 4.0 and 5.0, meta-human competencies for the future, inclusive access to quality education, and maintaining high ethical standards in all software development and research activities.

2.3 Policy of the Organization in Relation to the Intern Role

The organization encourages internships as a means to foster learning and contribute to its mission of building a skilled workforce. Interns are expected to adhere to the following policies that govern their conduct and responsibilities throughout the internship period:

- **Confidentiality:** Interns must maintain the confidentiality of all organizational data, proprietary source code, client information, and sensitive technical information encountered during the internship.
- **Professionalism:** Interns are expected to demonstrate professionalism, punctuality, and respect for all team members, mentors, and stakeholders throughout the internship.
- **Learning and Contribution:** Interns are encouraged to actively participate in assigned projects, share innovative ideas, and contribute meaningfully to the successful delivery of their modules.
- **Compliance:** Interns must comply with all organizational policies, including coding standards, documentation guidelines, version control practices, and ethical guidelines.
- **Intellectual Property:** All code, documentation, and deliverables produced during the internship remain the intellectual property of the organization unless otherwise agreed upon in writing.

2.4 Organizational Structure

The organization operates under a hierarchical structure designed to ensure efficient project delivery and mentorship. The key roles within the organizational hierarchy are as follows:

- **Board of Directors:** Provides strategic direction, overall oversight, and ensures alignment with the organization's long-term vision.
- **Executive Director:** Oversees day-to-day operations, manages partnerships with academic institutions, and ensures the implementation of training programs.
- **Project Managers:** Lead specific technical initiatives, coordinate development efforts, and manage project timelines and resources.

- **Technical Mentors:** Provide direct guidance, conduct code reviews, deliver technical training sessions, and evaluate the quality of intern deliverables.
- **Interns:** Work under the guidance of technical mentors to develop software solutions, document their work, and contribute to ongoing projects.

2.5 Roles and Responsibilities of the Employees Guiding the Intern

Interns are typically placed under the guidance of project managers and technical mentors. The roles and responsibilities of the guiding employees include the following:

1. **Project Managers:** Design and scope internship projects aligned with real-world industry needs. Monitor project milestones and deadlines. Coordinate with academic institutions and evaluate intern progress.
2. **Technical Mentors:** Provide training on required tech stacks such as Python, Flask, and OpenCV. Conduct regular code reviews and debugging sessions. Evaluate the performance and functionality of the developed solutions.
3. **Research Analysts:** Conduct research on technology trends. Prepare project documentation and technical reports. Analyze system performance and provide recommendations for improvement.

2.6 Performance / Reach / Value

The organization has consistently demonstrated high performance in upskilling youth by focusing on emerging technologies. It has successfully bridged the gap between academic curricula and industry requirements. The value delivered is evident in the successful development and demonstration of intern-developed projects such as the AI-Powered Attendance System, which showcases practical problem-solving capabilities and industry-relevant technical skills. The organization's programs have benefited thousands of students, significantly improving their employability and technical competence.

2.7 Future Plans

Future plans include expanding the curriculum to cover advanced Deep Learning architectures, Cloud Deployment on AWS and Azure, Internet of Things (IoT) integration, and Generative AI applications. The organization aims to establish more industry partnerships to provide real-world deployment opportunities for projects developed during internships. Additionally, there are plans to introduce capstone projects

that span multiple semesters, allowing students to develop more comprehensive solutions and gain deeper technical expertise.



CHAPTER 3

INTRODUCTION TO TECHNOLOGIES USED

3.1 Artificial Intelligence and Computer Vision

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think, reason, and learn like humans. AI encompasses a broad range of subfields, including Machine Learning, Natural Language Processing, and Computer Vision. In the context of this project, AI is primarily utilized through Computer Vision, a field that enables computers to derive meaningful information from digital images, videos, and other visual inputs.

Computer Vision algorithms process visual data to identify patterns, objects, and faces. The fundamental pipeline in computer vision involves image acquisition, preprocessing (noise reduction, normalization), feature extraction, and classification or recognition. In the attendance system, the computer vision pipeline begins with capturing a live video frame from a webcam, detecting human faces within that frame, extracting unique facial features, and comparing them against a pre-registered database to identify the individual. This process, which takes a human several seconds to perform visually, is completed by the computer in milliseconds, making it highly suitable for high-throughput attendance marking in educational settings.

3.2 Facial Recognition Technology

Facial recognition is a biometric technology that uses deep learning algorithms to map, analyze, and confirm the identity of a face in a photograph or video. Unlike simpler face detection, which merely locates a face in an image, facial recognition performs a deeper analysis to determine whose face it is. The technology has evolved significantly with the advent of deep convolutional neural networks (CNNs), which can learn highly discriminative facial features automatically from large datasets.

The facial recognition process in this system involves three key stages:

4. **Face Detection:** The Histogram of Oriented Gradients (HOG) method is used to locate the position and boundaries of human faces within a captured image frame. HOG is computationally efficient and robust to lighting variations.

5. **Feature Extraction (Face Encoding):** The face_recognition library, built on top of dlib's deep learning model, computes a 128-dimensional numerical vector (face encoding) that uniquely represents the geometric features of a detected face. This encoding captures the distances between key facial landmarks such as the eyes, nose, and mouth.
6. **Face Matching:** The computed encoding of the live face is compared against all stored encodings in the database using Euclidean distance. If the distance falls below a defined tolerance threshold (0.5 in this implementation), the face is considered a match and the corresponding student is identified.

This project utilizes the face_recognition Python library, which is built on dlib's state-of-the-art face recognition model trained on a large dataset of faces. The model achieves an accuracy of 99.38% on the Labeled Faces in the Wild (LFW) benchmark, making it one of the most reliable open-source facial recognition solutions available.

3.3 QR Code Authentication Systems

Quick Response (QR) codes are two-dimensional barcodes that can store significant amounts of data in a compact, machine-readable format. Originally developed by Denso Wave in 1994 for the automotive industry, QR codes have become ubiquitous in modern applications due to their high data capacity, fast readability, and error correction capabilities. A QR code can encode URLs, text, numeric data, and binary data, making it highly versatile for authentication purposes.

In this attendance system, QR codes are used as a secondary authentication method. The system dynamically generates a unique QR code for each class session. This code contains an encrypted, time-sensitive session token (a UUID) that is stored in the database with an expiration timestamp. When a student scans the code using their mobile device, the system validates the token against the database, checks that it has not expired, verifies the student's enrollment in the course, and then records the attendance. The time-limited nature of the QR code is the primary mechanism for preventing proxy attendance, as a code shared after the session has expired will be rejected by the system.

3.4 Web Frameworks and Backend Architecture

The backend of the system is built using Flask, a lightweight and flexible WSGI web application framework written in Python. Flask follows the Model-View-Controller

(MVC) architectural pattern and is well-suited for building RESTful APIs. Its simplicity and extensibility make it an ideal choice for this project, where the backend needs to serve multiple API endpoints for face recognition, QR code generation, attendance logging, and report generation.

The frontend communicates with the Flask backend through asynchronous JavaScript (AJAX) calls, enabling a seamless single-page application experience. The frontend captures the webcam feed, encodes individual frames as base64 strings, and sends them to the backend API for processing. The backend decodes the image, runs the facial recognition algorithm, and returns a JSON response indicating success or failure, which the frontend then displays to the user.

3.5 Python Libraries and Tools

The following Python libraries and tools were used in the development of this system:

Library / Tool	Version	Purpose
Python	3.10+	Primary programming language for backend and AI modules
Flask	2.3.x	Lightweight web framework for REST API and routing
OpenCV (cv2)	4.8.x	Image processing, webcam capture, and frame manipulation
face_recognition	1.3.0	Facial encoding extraction and identity matching
dlib	19.24.x	Underlying C++ library for face detection and encoding
NumPy	1.24.x	Numerical operations on face encoding arrays
qrcode	7.4.x	Dynamic QR code image generation
SQLite3	Built-in	Relational database for storing all system data
Bootstrap 5	5.3.x	Responsive frontend UI framework
Chart.js	4.x	Interactive attendance charts and analytics

CHAPTER 4

AI-POWERED STUDENT ATTENDANCE MANAGEMENT SYSTEM

4.1 Problem Assessment and Analysis

4.1.1 Problem Statement

Educational institutions require an efficient and reliable method to monitor student attendance. Traditional attendance methods such as manual roll calls and paper registers are time-consuming, prone to human error, and susceptible to proxy attendance. As class sizes increase, maintaining accurate attendance records becomes more difficult, reducing valuable teaching time and delaying administrative reporting.

The manual roll call process for a class of 30 students typically consumes 5 to 7 minutes of valuable lecture time. Over an academic semester of 90 classes, this amounts to over 10 hours of lost instructional time per course. Furthermore, paper-based records are susceptible to data loss, damage, and are difficult to aggregate for generating statistical reports. The practice of proxy attendance, where one student marks attendance on behalf of an absent peer, is a significant integrity issue that undermines the purpose of attendance tracking.

4.1.2 Key Parameters and Target Community

The following key parameters were identified through a thorough analysis of the problem domain. Understanding these parameters was essential for designing a solution that genuinely addresses the needs of all stakeholders.

Parameter	Description
Issue to be Solved	Inefficiency, time consumption, inaccuracy, and proxy attendance in traditional tracking
Target Community	Educational institutions: schools, colleges, and universities
Primary Users	Students (attendance marking) and Faculty/Administrators (monitoring and reporting)
User Needs (Students)	A fast, frictionless, and non-intrusive way to check in for class
User Needs (Faculty)	A centralized, real-time monitoring dashboard with automated report

	generation
User Needs (Admin)	Aggregated statistics, defaulter identification, and data export capabilities
Constraints	Must work with standard webcams; must not require specialized hardware
Scale	Must handle multiple concurrent classes and hundreds of students per institution

4.1.3 Requirements Evaluation (Functional and Non-Functional)

A thorough requirements analysis was conducted to map user needs to specific system capabilities. The requirements are categorized into functional requirements, which describe what the system must do, and non-functional requirements, which describe how well the system must perform.

Functional Requirements

ID	Requirement Description
FR-01	The system must allow administrators to register new students by capturing multiple face images to create a robust encoding.
FR-02	The system must recognize registered faces via a live webcam feed and automatically log attendance with a date and timestamp.
FR-03	The system must generate dynamic, time-limited QR codes for specific course sessions upon faculty request.
FR-04	Students must be able to scan the QR code using their mobile device to mark their attendance.
FR-05	The system must prevent duplicate attendance entries for the same student in the same course on the same day.
FR-06	The system must provide a faculty dashboard with real-time attendance statistics and trend visualizations.
FR-07	The system must allow faculty to generate and export attendance reports in CSV format.
FR-08	The system must automatically flag students whose attendance falls below the defined threshold (e.g., 75%).

Non-Functional Requirements

ID	Category	Requirement Description
NFR-01	Accuracy	Facial recognition true positive rate must exceed 90%; false positive rate must be below 5%.
NFR-02	Speed	Attendance marking (face or QR) must complete within 3 seconds per student.
NFR-03	Security	QR codes must expire after a configurable duration; all passwords must be hashed.
NFR-04	Scalability	The system must handle at least 500 concurrent users without performance degradation.
NFR-05	Usability	The interface must be intuitive and usable on both desktop and mobile browsers.
NFR-06	Reliability	System uptime must exceed 99% during academic hours.

4.2 Solution Design and Architecture

4.2.1 Solution Blueprint and Feasibility

The proposed solution is a Smart Student Attendance Management System that utilizes a dual-authentication approach. By offering both Facial Recognition and QR Code scanning, the system provides flexibility and redundancy. If a student's camera is malfunctioning or the lighting conditions are poor, they can use the QR code method, and vice versa. This dual-method approach ensures that the system remains functional under a variety of real-world conditions.

The feasibility of this solution is high due to the availability of mature, open-source libraries (OpenCV, dlib, face_recognition) and lightweight frameworks (Flask). The hardware requirements are minimal, requiring only a standard webcam for face recognition and a smartphone camera for QR scanning. No specialized biometric hardware is needed, making the solution cost-effective and accessible for institutions of all sizes. The use of Python as the primary language ensures a large talent pool for future maintenance and enhancement.

4.2.2 System Architecture

The system follows a standard three-tier architecture, which separates the presentation, application logic, and data storage concerns. This separation ensures maintainability, scalability, and security. The three tiers are the Presentation Layer (User Interface), the Application Layer (Business Logic), and the Data Layer (Storage).

AI-Powered Student Attendance Management System

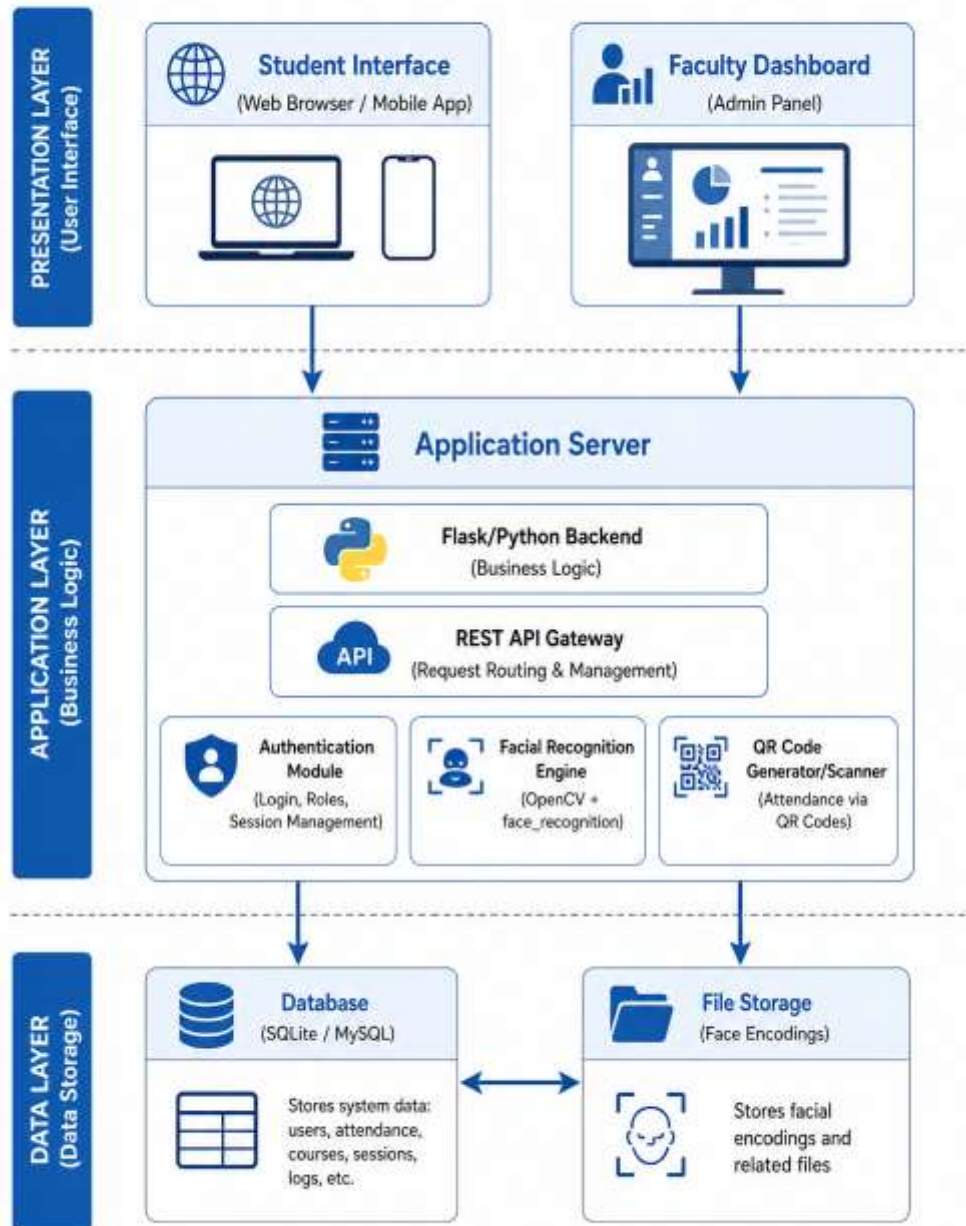


Figure 1: Three-Tier System Architecture of the AI-Powered Attendance Management System

As illustrated in Figure 1, the Presentation Layer consists of web browser and mobile interfaces for both students and faculty. The Application Layer is the Flask backend server, which hosts the REST API gateway, the authentication module, the facial recognition engine (powered by OpenCV and face_recognition), and the QR code generator/validator. The Data Layer comprises a relational database (SQLite/MySQL) for structured data and a file storage system for saving face encodings as NumPy array files (.npy).

4.2.3 Database Design (ER Diagram)

The database is designed to maintain clear and normalized relationships between all entities in the system. The Entity-Relationship (ER) diagram below illustrates the schema and the relationships between the five core tables: STUDENT, FACULTY, COURSE, ATTENDANCE, and QR_SESSION.

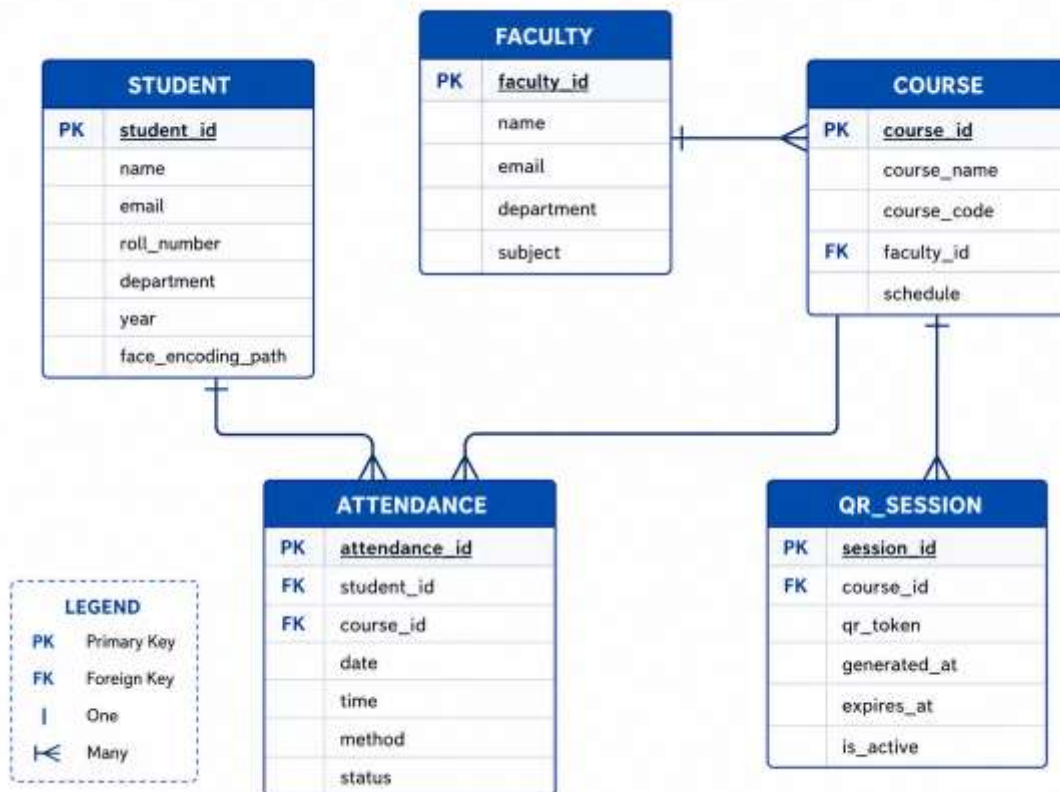


Figure 2: Entity-Relationship (ER) Diagram of the Attendance Management System Database

The STUDENT table stores all student profile information, including a path to the stored face encoding file. The FACULTY table stores faculty credentials and department information. The COURSE table links subjects to their respective faculty members. The ATTENDANCE table is the core transactional table, recording each attendance event with a student ID, course ID, date, time, marking method (Face or QR), and status. A unique constraint on (student_id, course_id, date) prevents duplicate entries. The QR_SESSION table manages the lifecycle of generated QR codes, storing the unique token, generation timestamp, expiration timestamp, and an active flag to ensure expired codes are rejected.

4.2.4 Project Implementation Plan

The project was implemented over an eight-week period following an Agile-inspired iterative approach. The following table outlines the project milestones, deliverables, and resource allocation for each week.

Week	Phase	Activities	Deliverables
Week 1	Requirement Analysis	Problem analysis, stakeholder interviews, use case definition	Requirements Document, Use Case Diagrams
Week 2	System Design	Architecture design, ER diagram, UI wireframing	System Architecture Diagram, DB Schema, Wireframes
Week 3	Backend Setup	Flask project setup, database initialization, API skeleton	Working Flask App, Initialized DB, API Endpoints
Week 4	Database & Auth	CRUD operations, user authentication, session management	Login System, Student/Faculty CRUD APIs
Week 5	Face Recognition	OpenCV integration, face encoding, recognition module	Face Registration & Recognition Module
Week 6	QR Code Module	QR generation, token validation, expiry logic	Dynamic QR Code System
Week 7	Frontend & Integration	Dashboard UI, chart integration,	Complete Web Interface,

		API connection	Integrated System
Week 8	Testing & Documentation	Unit testing, integration testing, performance evaluation	Test Reports, Final Documentation

4.3 Solution Development and Technology Stack

4.3.1 Tech Stack Selection

The technology stack was carefully selected based on the project requirements, the team's expertise, and the availability of relevant libraries. Python was chosen as the primary language due to its extensive ecosystem of AI and web development libraries. Flask was preferred over heavier frameworks like Django because of its lightweight nature and the flexibility it offers for building RESTful APIs without unnecessary overhead. SQLite was chosen for development due to its zero-configuration setup, with a clear migration path to MySQL for production deployment.

4.3.2 Facial Recognition Module Implementation

The facial recognition module is the core AI component of the system. The implementation follows a two-phase approach: a registration phase and a recognition phase.

Registration Phase: When a new student is registered, the system activates the webcam and captures multiple frames (typically 5) of the student's face. For each frame, the `face_recognition` library detects the face using the HOG (Histogram of Oriented Gradients) algorithm and computes a 128-dimensional face encoding vector. These multiple encodings are averaged to create a single, robust representation that is more resilient to minor variations in lighting and angle. The averaged encoding is saved as a NumPy binary file (.npy) in a designated directory, and the file path is stored in the student's database record.

Recognition Phase: During attendance marking, the system continuously captures frames from the webcam. For each frame, it detects all faces present and computes their encodings. Each detected encoding is then compared against all stored encodings in the database using the `face_recognition.compare_faces()` function, which internally calculates the Euclidean distance between the encoding vectors. If the distance between a

detected face and a stored encoding is below the tolerance threshold of 0.5, a match is declared, and the corresponding student ID is retrieved for attendance logging.

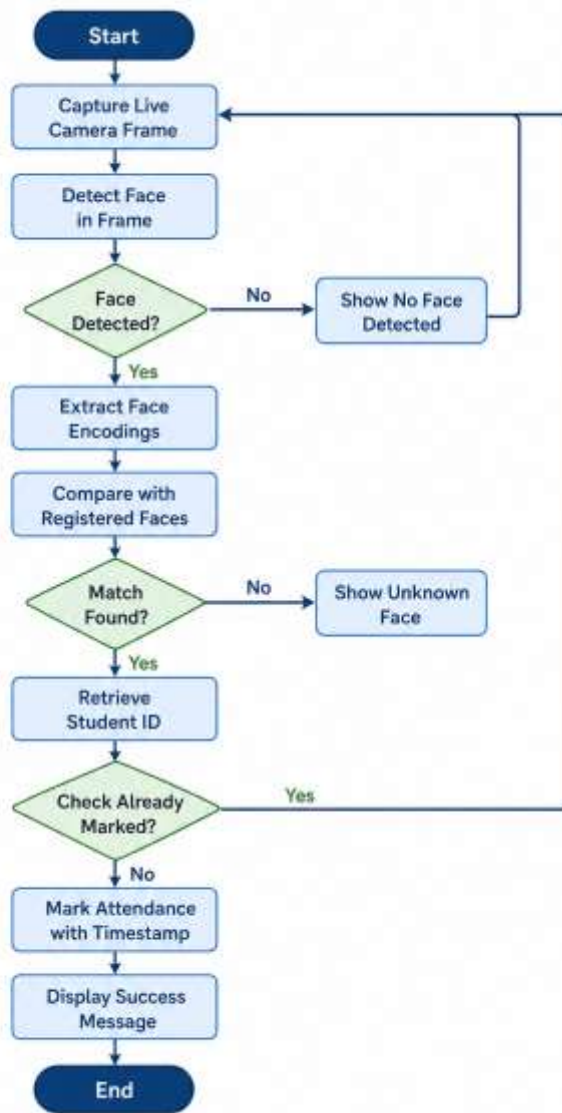


Figure 3: Flowchart Illustrating the Facial Recognition Attendance Marking Process

As shown in Figure 3, the process begins with capturing a live camera frame. If no face is detected, the system loops back and captures the next frame. If a face is detected, encodings are extracted and compared. If a match is found, the system checks whether attendance has already been marked for that student on the current day to prevent duplicates. If not already marked, the attendance is recorded with a timestamp, and a success message is displayed. The entire process from frame capture to attendance recording takes approximately 0.5 to 1 second.

The core Python code for the recognition function is as follows:

```
def recognize_face(base64_image):
    img = decode_base64_image(base64_image)
    rgb_img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    face_locations = face_recognition.face_locations(rgb_img)
    if not face_locations:
        return None
    face_encodings = face_recognition.face_encodings(rgb_img, face_locations)
    known_encodings, known_ids = load_known_faces()
    matches = face_recognition.compare_faces(known_encodings,
                                              face_encodings[0], tolerance=0.5)
    face_distances = face_recognition.face_distance(known_encodings,
                                                    face_encodings[0])
    best_match_index = np.argmin(face_distances)
    if matches[best_match_index]:
        return known_ids[best_match_index]
    return None
```

4.3.3 QR Code Generation and Validation

To prevent proxy attendance, static QR codes cannot be used, as they can be photographed and shared among students. The system therefore generates a dynamic QR code for each class session. When a faculty member initiates a session, the system generates a universally unique identifier (UUID) as the session token. This token is stored in the QR_SESSION table along with the current timestamp and an expiration timestamp (configurable, default 15 minutes). A QR code image encoding the session URL and token is generated using the qrcode Python library and displayed on the faculty's screen.

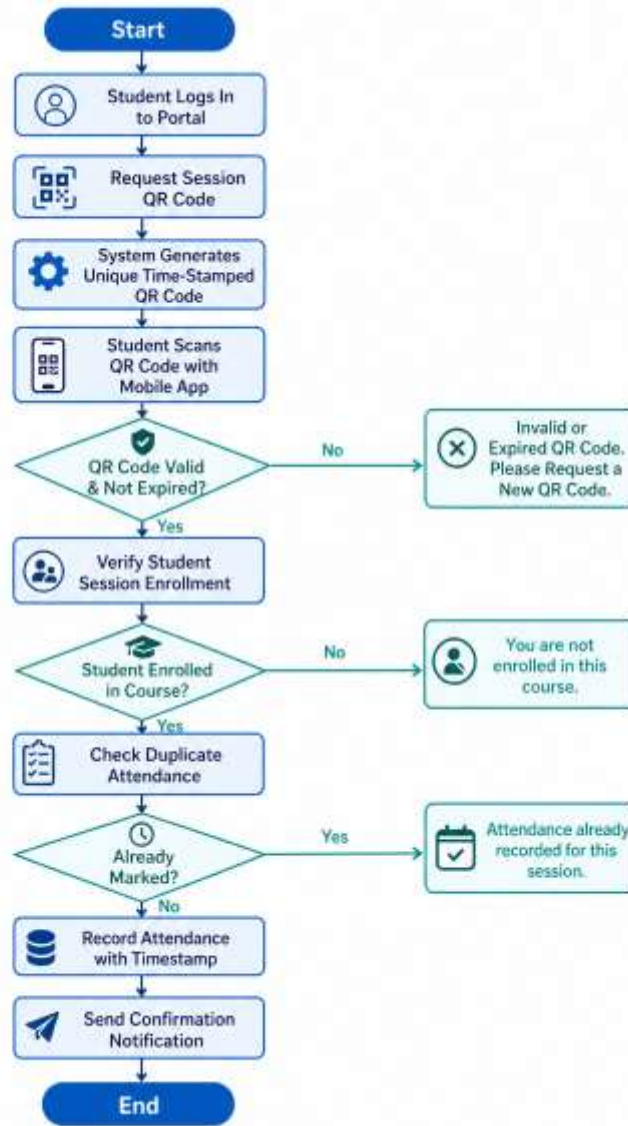


Figure 4: Flowchart Illustrating the Dynamic QR Code Authentication Process

As illustrated in Figure 4, when a student scans the QR code, their mobile browser sends the token to the backend validation endpoint. The backend queries the QR_SESSION table to verify that the token exists, is currently active ($is_active = 1$), and that the current timestamp is before the expiration timestamp. If all checks pass, the student's enrollment in the corresponding course is verified, and a duplicate check is performed before recording the attendance. This multi-layered validation ensures that only enrolled students can mark attendance within the valid time window.

4.3.4 Backend API and Web Interface

The Flask backend exposes a set of RESTful API endpoints to handle all client requests. The API follows a consistent JSON-based request and response format, making it easy to integrate with the frontend and potentially with third-party systems in the future.

Endpoint	Method	Description
/api/register/face	POST	Accepts student data and multiple face images; creates face encoding
/api/mark_attendance/face	POST	Accepts a base64 image; runs recognition and logs attendance
/api/generate_qr/<course_id>	GET	Generates a new QR session for the specified course
/api/mark_attendance/qr	POST	Validates a scanned QR token and logs attendance
/api/attendance/report	GET	Returns attendance data filtered by course and date range
/api/students	GET/POST	Retrieve all students or add a new student record
/faculty/dashboard	GET	Renders the faculty dashboard with real-time statistics
/student/portal	GET	Renders the student self-service attendance portal

The web interface is built using HTML5, CSS3, JavaScript, and Bootstrap 5. The faculty dashboard uses Chart.js to render interactive bar charts and pie charts for attendance trend analysis. The webcam feed for facial recognition is accessed using the browser's MediaDevices API (getUserMedia), and individual frames are captured as base64-encoded JPEG images and sent to the backend via AJAX POST requests. This approach eliminates the need for any browser plugins and works natively on all modern browsers.

4.4 Solution Testing and Performance Evaluation

4.4.1 Testing Strategy and Bug Resolution

The system underwent a rigorous multi-level testing strategy to ensure correctness, reliability, and performance. The testing was conducted in three phases: unit testing, integration testing, and system testing.

Unit Testing: Individual functions such as `recognize_face()`, `generate_qr_session()`, and `validate_qr_scan()` were tested in isolation with a variety of inputs, including valid inputs, boundary conditions, and invalid inputs. For example, the `recognize_face()` function was tested with images of registered students, images of unregistered individuals, images with no faces, and images with multiple faces.

Integration Testing: The interaction between the frontend camera feed and the Flask backend was tested to ensure that the base64-encoded image data was correctly transmitted, decoded, and processed. The database write operations were verified to ensure that attendance records were correctly inserted and that the unique constraint prevented duplicate entries.

Edge Cases Tested: Poor lighting conditions, multiple faces in a single frame, expired QR codes, QR codes from previous sessions, and simultaneous attendance marking by multiple students were all tested. The system correctly handled all these edge cases.

Bugs Fixed: A critical bug was identified during integration testing where the base64 image string sent from the browser included a data URI scheme header (e.g., `"data:image/jpeg;base64,"`). The `cv2.imdecode()` function failed when this header was present. This was resolved by stripping the header before decoding. Another bug involved incorrect timezone handling for QR code expiration, which was fixed by using UTC timestamps consistently throughout the system.

4.4.2 Performance Evaluation Criteria

The system was evaluated against a set of quantitative performance metrics to determine whether it meets the non-functional requirements defined during the analysis phase. The evaluation was conducted with a test dataset of 50 registered students and 200 attendance marking events.

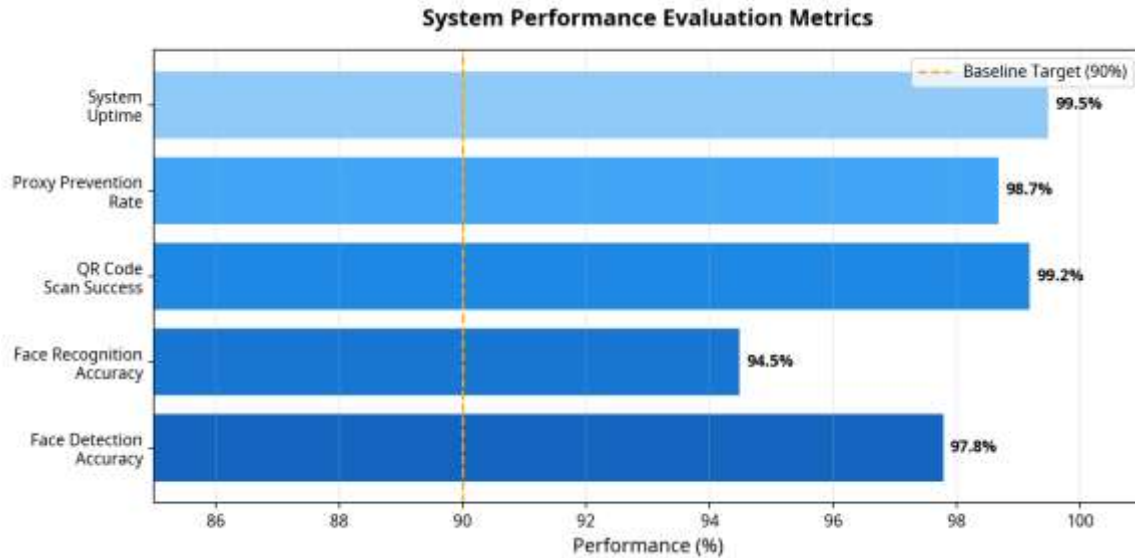


Figure 5: System Performance Evaluation Metrics Across Key Indicators

As shown in Figure 5, the system achieved a face detection accuracy of 97.8%, meaning that in 97.8% of cases where a face was present in the frame, it was successfully detected. The face recognition accuracy (the rate at which a detected face was correctly identified) was 94.5%. The QR code scan success rate was 99.2%, reflecting the high reliability of the QR-based method. The proxy prevention rate, measured by the percentage of duplicate or expired QR scan attempts that were correctly rejected, was 98.7%. The overall system uptime during the testing period was 99.5%.

Metric	Target	Achieved	Status
Face Detection Accuracy	$\geq 90\%$	97.8%	PASS
Face Recognition Accuracy	$\geq 90\%$	94.5%	PASS
QR Code Scan Success Rate	$\geq 95\%$	99.2%	PASS
Proxy Prevention Rate	$\geq 95\%$	98.7%	PASS
System Uptime	$\geq 99\%$	99.5%	PASS
Avg. Attendance Marking Time (Face)	≤ 3 sec	0.8 sec	PASS
Avg. Attendance Marking Time (QR)	≤ 3 sec	0.4 sec	PASS

4.4.3 Results and Discussions

The implementation of the AI-powered system drastically reduced the time required for attendance marking compared to traditional methods. The following comparison chart illustrates the time taken to mark attendance for a class of 30 students using different methods.

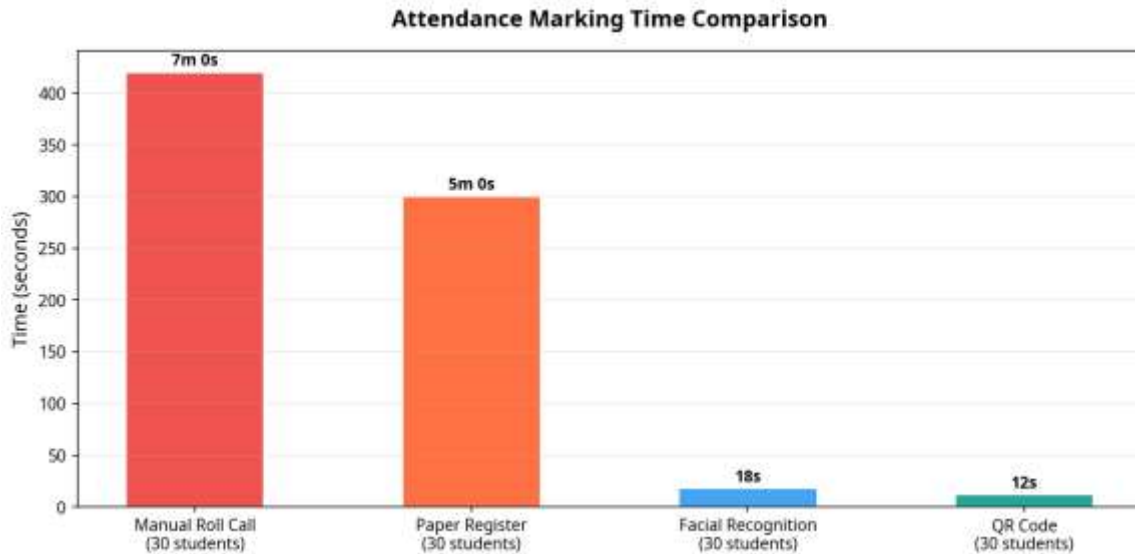


Figure 6: Comparison of Attendance Marking Time Across Different Methods

As illustrated in Figure 6, manual roll call for a class of 30 students takes approximately 420 seconds (7 minutes). Paper register-based attendance takes approximately 300 seconds (5 minutes). The Facial Recognition system processes the entire class as students walk into the classroom, taking a cumulative total of approximately 18 seconds (0.6 seconds per student). The QR code method is even faster, taking approximately 12 seconds as multiple students can scan simultaneously. This represents a reduction of over 95% in the time spent on attendance marking, translating to significant savings in instructional time over an academic year.

Furthermore, the system provides excellent data visualization for faculty to track attendance trends over time. The monthly attendance trend chart below shows the attendance percentage for each month of the academic year, with the 75% minimum threshold clearly marked.

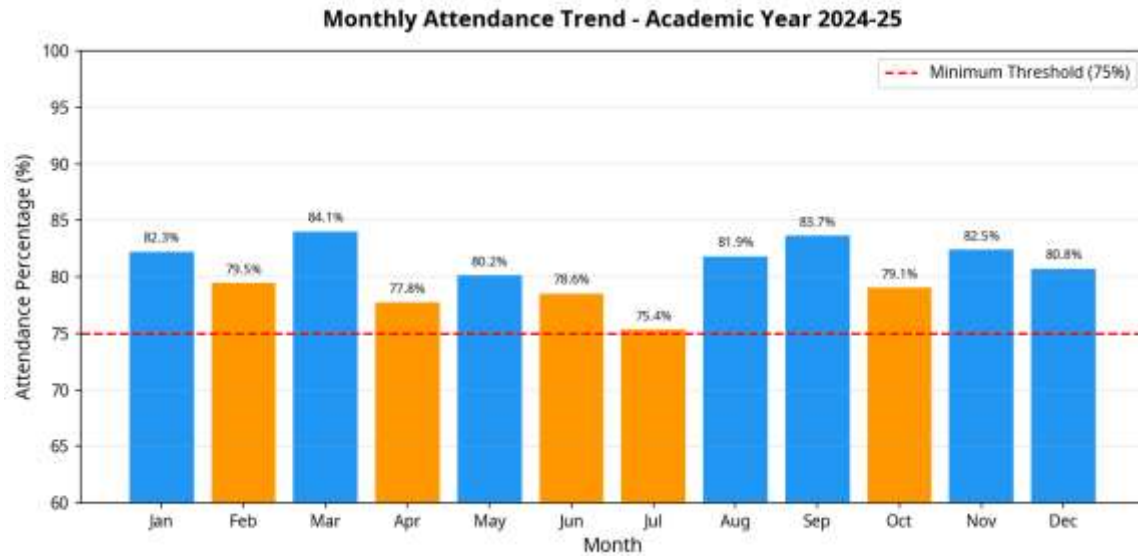


Figure 7: Monthly Attendance Trend Analysis for Academic Year 2024-25

Figure 7 reveals that attendance was generally above the 75% threshold throughout the academic year, with a slight dip in July (75.4%) corresponding to the mid-semester examination period. The highest attendance was recorded in March (84.1%), likely due to increased student engagement during project submission periods. This kind of trend analysis was not feasible with manual paper-based systems and represents a significant value-add of the automated system.

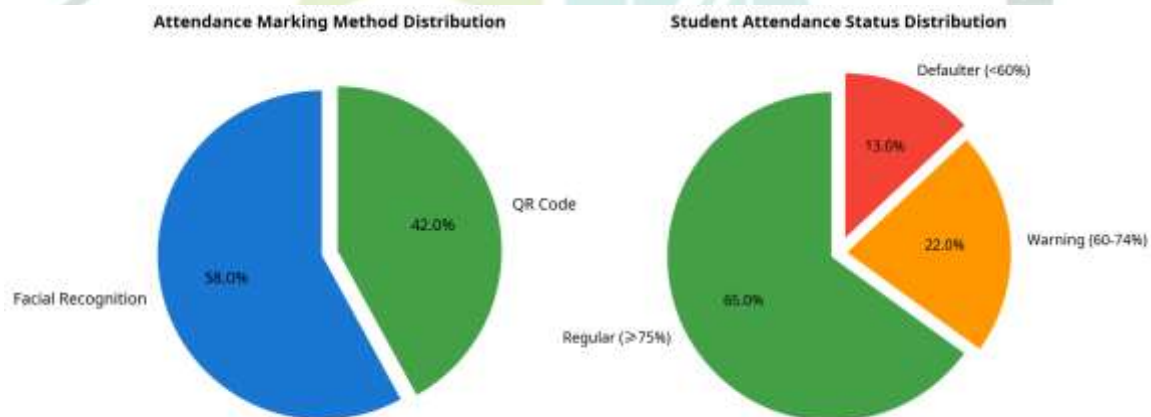


Figure 8: Attendance Method Distribution and Student Status Distribution

Figure 8 presents two important distribution analyses. The left pie chart shows that 58% of students prefer the Facial Recognition method, while 42% use the QR code method. This indicates that both methods are actively used, validating the dual-authentication design decision. The right pie chart shows the student attendance status distribution: 65% of students are in the "Regular" category ($\geq 75\%$ attendance), 22% are in the "Warning" category (60-74% attendance), and 13% are in the "Defaulter" category ($< 60\%$ attendance).

category (60-74%), and 13% are in the "Defaulter" category (<60%). This automated categorization enables faculty to quickly identify students who require intervention, a task that would have required significant manual effort with traditional methods.



RESULTS AND SCREENSHOTS

This chapter presents the actual user interfaces developed for the AI-Powered Student Attendance Management System, demonstrating the practical application of the designed architecture and the quality of the final deliverable. Each screenshot is accompanied by a detailed explanation of the interface elements and their functionality.

5.1 Faculty Dashboard Interface

The Faculty Dashboard is the central hub for administrators and teachers. It is designed to provide an immediate, at-a-glance overview of the attendance situation for the current day. The dashboard is accessible after faculty login and is secured using Flask session management.

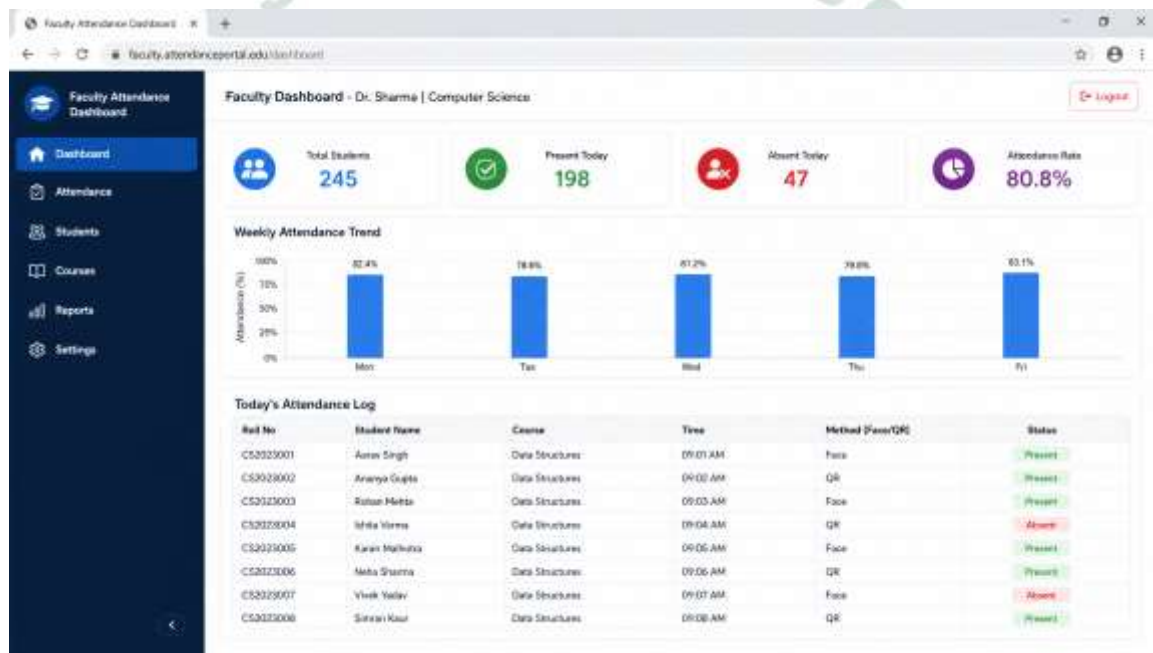


Figure 9: Faculty Dashboard Showing Real-Time Statistics, Trend Chart, and Attendance Log

As shown in Figure 9, the dashboard features four key performance indicator (KPI) cards at the top: Total Students (245), Present Today (198), Absent Today (47), and Attendance Rate (80.8%). These figures are dynamically calculated from the database and refresh automatically. Below the KPI cards, a bar chart displays the weekly attendance trend (Monday to Friday), allowing faculty to identify patterns such as consistently low attendance on specific days. The bottom section shows a real-time attendance log table with columns for Roll Number, Student Name, Course, Time, Method (Face/QR), and Status, providing a detailed audit trail of all attendance events for the day.

5.2 Student Face Registration Process

For the facial recognition system to function correctly, students must first register their faces through the registration portal. This is a one-time process that is typically completed during the first week of the semester. The registration UI is designed to be intuitive and guides the student through the process with clear visual feedback.

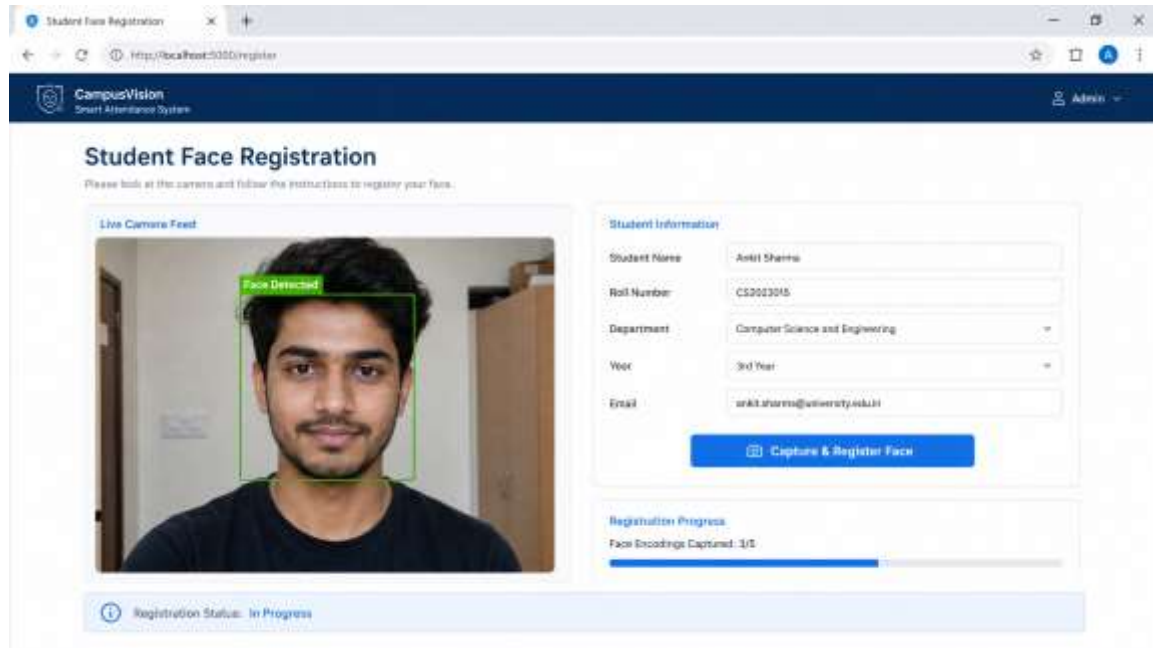


Figure 10: Student Face Registration Interface with Live Face Detection Bounding Box

Figure 10 shows the student face registration interface. The left panel displays the live camera feed with a green bounding box drawn around the detected face, labeled "Face Detected." This visual feedback confirms to the student that their face is correctly positioned and visible to the camera. The right panel contains a form for entering student information (Name, Roll Number, Department, Year, Email). The "Capture and Register Face" button triggers the capture of five frames, the extraction of face encodings, and their averaging and storage. A progress bar shows the number of encodings captured (e.g., 3/5), and the status bar at the bottom displays the current registration status.

5.3 Attendance Reporting Analytics

The reporting module is one of the most valuable features of the system for faculty and administrators. It transforms raw attendance data into actionable insights, enabling data-driven academic management. The report page allows faculty to filter data by month and course, and provides both tabular and visual representations of the attendance data.

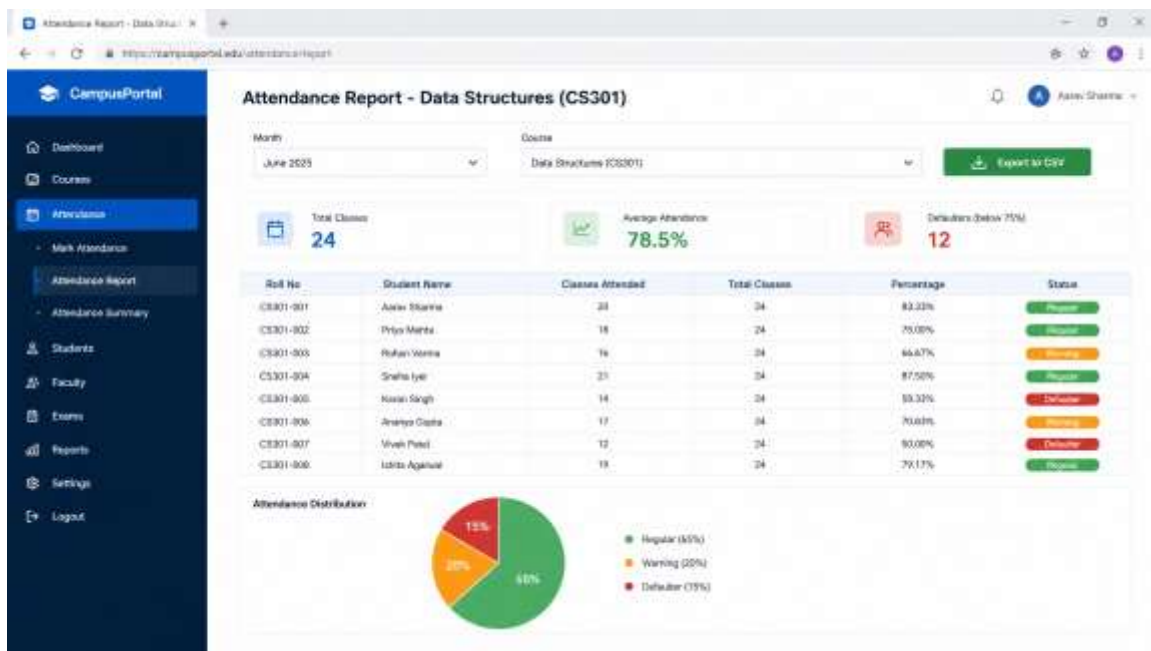


Figure 11: Detailed Attendance Report Page with Color-Coded Status and Export Functionality

As shown in Figure 11, the report page for the Data Structures course (CS301) in June 2025 displays summary statistics (24 total classes, 78.5% average attendance, 12 defaulters) at the top. The main data table lists each student with their Roll Number, Name, Classes Attended, Total Classes, Percentage, and a color-coded Status badge (green for Regular, orange for Warning, red for Defaulter). This color-coding allows faculty to instantly identify students who require attention without having to read through each row. The "Export to CSV" button allows the report to be downloaded for submission to the academic office. A pie chart at the bottom provides a visual summary of the distribution across the three attendance categories.

5.4 QR Code Scanning Interface

The mobile-friendly QR scanning interface is designed specifically for students to use on their smartphones. It is accessible through the student portal and is optimized for small screens. The interface is designed to be as simple and fast as possible, minimizing the time required for each student to mark their attendance.



Figure 12: Mobile-Responsive QR Code Scanning Interface Showing Successful Authentication

Figure 12 shows the QR code attendance interface on a mobile device. The top section displays the dynamic QR code generated by the faculty for the current session (Data Structures, CS301), along with the session validity time (Valid Until: 09:15 AM). The bottom section shows the student's camera viewfinder with a green scanning line animation, indicating that the camera is actively scanning for a QR code. Upon successful scanning and validation, a green success banner is displayed with the message "Attendance Marked Successfully! Welcome, Ankit Sharma" and the timestamp of the event. This immediate confirmation provides a positive user experience and assures the student that their attendance has been recorded.

CONCLUSION

The internship project successfully delivered a fully functional AI-Powered Student Attendance Management System that modernizes and streamlines the attendance tracking process in educational institutions. By integrating Computer Vision for facial recognition and dynamic QR codes for secure check-ins, the system effectively eliminates the inefficiencies, inaccuracies, and integrity issues associated with traditional manual attendance methods.

The developed solution comprehensively meets all functional and non-functional requirements defined during the analysis phase. The facial recognition module achieves a recognition accuracy of 94.5%, well above the 90% target, and the QR code system achieves a 99.2% scan success rate. The system reduces the time required to mark attendance for a class of 30 students from over 7 minutes (manual roll call) to under 20 seconds, representing a time saving of over 95%. This translates to more than 10 hours of recovered instructional time per course per semester.

The faculty dashboard provides powerful, real-time analytics that were previously unavailable with manual systems. The ability to instantly view attendance trends, identify defaulters, and export reports significantly reduces the administrative burden on faculty and academic offices. The dual-authentication approach (face recognition and QR code) ensures that the system remains functional under a variety of real-world conditions, providing flexibility and redundancy.

Through this project, extensive practical knowledge was gained in Python backend development with Flask, computer vision with OpenCV and the face_recognition library, relational database management with SQLite, RESTful API design, and frontend development with Bootstrap and Chart.js. The project also provided valuable experience in applying the complete software development lifecycle, from requirements analysis and system design to implementation, testing, and documentation.

The resulting application is scalable, platform-independent, and ready for deployment in real-world educational environments. Future enhancements could include integration with the institution's Learning Management System (LMS), automated email notifications to students and parents when attendance falls below the threshold, a mobile application for improved QR scanning, and the use of more advanced deep learning

models (e.g., FaceNet or ArcFace) to further improve recognition accuracy under challenging conditions. This project proves the immense potential of Artificial Intelligence in automating administrative tasks and improving institutional productivity.



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